

ELEVATE LABS

TASK 16

Incident Response & Security Breach Simulation

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Domain: Cybersecurity

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1. Simulate a basic security incident such as repeated failed logins or unauthorized access.

A security incident is any event that threatens the confidentiality, integrity, or availability of a system.

One common basic incident is repeated failed login attempts.

This usually indicates a brute-force attack where an attacker tries multiple passwords.

If the password is weak, the attacker may eventually gain access.

Another common incident is unauthorized access attempts.

This happens when a user tries to access files or systems without proper permission.

Such actions may indicate an insider threat or external hacking attempt.

These activities are usually recorded in authentication and system logs.

Monitoring logs helps detect suspicious behavior early.

Simulating these incidents in a lab helps understand detection and response techniques.

Step 2: Generate Failed Login Attempts

Run:

ssh localhost

Enter a **wrong password** 3–5 times.

```
(kali㉿kali)-[~]
$ ssh localhost
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Linux kali 6.18.9+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.9-1kali1 (2026-02-10) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

Step 3: View Authentication Logs

`sudo journalctl | grep "Failed password"`

```
(kali㉿kali)-[~]
$ sudo journalctl | grep "Failed password"

Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 10 09:18:42 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:49 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:55 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:41:09 kali sshd-session[51714]: Failed password for root from ::1 port 39470 ssh2
Feb 13 10:28:12 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:28:19 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:32:12 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:19 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:24 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
```

View SSH Authentication Logs

`sudo journalctl _COMM=sshd`

```
(kali㉿kali)-[~]
$ sudo journalctl _COMM=sshd

Feb 03 10:45:42 kali sshd[3506]: Server listening on 0.0.0.0 port 22.
Feb 03 10:45:42 kali sshd[3506]: Server listening on :: port 22.
-- Boot 57ac48aa1e4948a4bdeec1182c27938 --
Feb 10 09:10:35 kali sshd[23403]: Server listening on 0.0.0.0 port 22.
Feb 10 09:10:35 kali sshd[23403]: Server listening on :: port 22.
Feb 10 09:17:14 kali sshd[23403]: Received signal 15; terminating.
Feb 10 09:17:14 kali sshd[26853]: Server listening on 0.0.0.0 port 22.
Feb 10 09:17:14 kali sshd[26853]: Server listening on :: port 22.
Feb 10 09:43:05 kali sshd[26853]: Timeout before authentication for connection from ::1 to ::1, pid = 51714
-- Boot aa6982c5679e410e9d148b04c82e6ea2 --
Feb 12 08:36:29 kali sshd[852]: Server listening on 0.0.0.0 port 22.
Feb 12 08:36:29 kali sshd[852]: Server listening on :: port 22.
Feb 12 08:57:21 kali sshd[852]: Received signal 15; terminating.
-- Boot 8b3a4faee2d043a080454ddfbfc5e3f2 --
Feb 12 08:57:53 kali sshd[708]: Server listening on 0.0.0.0 port 22.
Feb 12 08:57:53 kali sshd[708]: Server listening on :: port 22.
Feb 12 09:09:44 kali sshd[708]: Received signal 15; terminating.
-- Boot 6bc73c9f35eb4c128217b0df4717cd79 --
Feb 12 09:11:14 kali sshd[734]: Server listening on 0.0.0.0 port 22.
Feb 12 09:11:14 kali sshd[734]: Server listening on :: port 22.
Feb 12 09:37:19 kali sshd[734]: Received signal 15; terminating.
-- Boot 45448d846c4f43de8f8b3d4dc7d9df96 --
```

Count Failed Attempts

`sudo journalctl | grep "Failed password" | wc -l`

```
(kali㉿kali)-[~]
$ sudo journalctl | grep "Failed password" | wc -l
10
```

2. Identify suspicious activity by analyzing system and authentication logs

- Identifying suspicious activity involves carefully analyzing system and authentication logs.

Logs record events such as logins, logouts, errors, and system changes.

Authentication logs specifically track successful and failed login attempts.

Multiple failed login attempts in a short time may indicate a brute-force attack.

Login attempts at unusual hours can also be suspicious.

Access from unknown or unfamiliar IP addresses is another warning sign.

Repeated attempts to access restricted files suggest unauthorized access.

Sudden privilege escalation, such as a normal user gaining admin rights, is suspicious.

Frequent account lockouts may indicate password-guessing attacks.

New user account creation without authorization should be investigated.

Changes to critical system files may indicate malicious activity.

Logs showing disabled security services are a serious red flag.

Comparing current log activity with normal behavior helps detect anomalies.

Security tools like SIEM systems help automate log analysis.

Regular log monitoring helps detect threats early and prevent system compromise.

View Authentication Failures

`sudo journalctl | grep "Failed"`

```
(kali㉿kali)-[~]
└─$ sudo journalctl | grep "Failed"
Nov 26 23:23:41 kali wireplumber[964]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:45 kali lightdm[930]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:23:46 kali lightdm[1034]: Failed to write utmpx: No such file or directory
Nov 26 23:23:46 kali wireplumber[1074]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:23 kali wireplumber[743]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:28 kali lightdm[707]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:25:28 kali lightdm[810]: Failed to write utmpx: No such file or directory
Nov 26 23:25:28 kali wireplumber[849]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 27 08:38:12 kali wireplumber[839]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 27 08:38:16 kali lightdm[804]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 27 08:38:17 kali lightdm[908]: Failed to write utmpx: No such file or directory
```

View SSH Login Activity

`sudo journalctl -u ssh`

```
(kali㉿kali)-[~]
└─$ sudo journalctl -u ssh
Feb 03 10:45:42 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 03 10:45:42 kali sshd[3506]: Server listening on 0.0.0.0 port 22.
Feb 03 10:45:42 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 03 10:45:42 kali sshd[3506]: Server listening on :: port 22.
Feb 03 10:46:46 kali sshd-session[3835]: Connection closed by 127.0.0.1 port 49538 [preauth]
Feb 03 10:47:43 kali sshd-session[4072]: Connection closed by 127.0.0.1 port 33820 [preauth]
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=kali
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request wbcbLoginUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'kali')
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 50610 sshd
```

Count Failed Login Attempts

`sudo journalctl | grep "Failed password" | wc -l`

```
(kali㉿kali)-[~]
└─$ sudo journalctl | grep "Failed password" | wc -l
10
```

Step 5: Identify Source IP Addresses

`sudo journalctl | grep "Failed password"`

```
(kali㉿kali)-[~]
$ sudo journalctl | grep "Failed password"
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 10 09:18:42 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:49 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:55 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:41:09 kali sshd-session[51714]: Failed password for root from ::1 port 39470 ssh2
Feb 13 10:28:12 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:28:19 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:32:12 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:19 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:24 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
```

Check Successful Logins (Compare)

`last`

```
(kali㉿kali)-[~]
$ last
kali      tty7          :0                Fri Feb 13 10:31 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:31 - 10:31 (00:00)
kali      pts/1         ::1               Fri Feb 13 10:28 - still logged in
kali      pts/1         ::1               Fri Feb 13 10:27 - 10:28 (00:00)
kali      tty7          :0                Fri Feb 13 10:27 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:26 - 10:27 (00:00)
kali      tty7          :0                Fri Feb 13 08:24 - 08:28 (00:04)
lightdm   tty7          :0                Fri Feb 13 08:24 - 08:24 (00:00)
kali      tty7          :0                Fri Feb 13 08:22 - still logged in
lightdm   tty7          :0                Fri Feb 13 08:22 - 08:22 (00:00)
kali      tty7          :0                Fri Feb 13 07:31 - still logged in
lightdm   tty7          :0                Fri Feb 13 07:31 - 07:31 (00:00)
kali      tty7          :0                Thu Feb 12 09:37 - 09:49 (00:11)
```

4. Classify the incident based on attack type and severity

The first step is to determine whether the incident is internal or external. If multiple failed logins are detected, it may be classified as a brute-force attack.

If malicious software is found, it is classified as a malware infection.

Unauthorized access to sensitive files may be classified as a privilege escalation attack.

Phishing-related access can be classified as a social engineering attack.

Next, the severity of the incident must be evaluated.

Low severity incidents have minimal impact and no data loss.

Medium severity incidents may affect limited systems or user accounts.

High severity incidents involve data breaches, system compromise, or

service disruption.

Critical severity incidents impact core systems or confidential data.

The number of affected users also influences severity classification.

Financial loss and reputational damage are considered during assessment.

Organizations often use an incident severity matrix for proper categorization.

Proper classification helps in deciding the response priority and resources needed.

View Failed Authentication Attempts

On modern Linux (Kali/Ubuntu):

`sudo journalctl | grep "Failed"`

```
(kali@kali)-[~]
└─$ sudo journalctl | grep "Failed"
Nov 26 23:23:41 kali wireplumber[964]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:45 kali lightdm[930]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:23:46 kali lightdm[1034]: Failed to write utmpx: No such file or directory
Nov 26 23:23:46 kali wireplumber[1074]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:23 kali wireplumber[743]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:28 kali lightdm[707]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:25:28 kali lightdm[810]: Failed to write utmpx: No such file or directory
Nov 26 23:25:28 kali wireplumber[849]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:28 kali wireplumber[839]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 27 08:38:12 kali lightdm[930]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
```

Identify Source and Username

`sudo journalctl | grep "Failed password"`

```
(kali@kali)-[~]
└─$ sudo journalctl | grep "Failed password"
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 10 09:18:42 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:49 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:55 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:41:09 kali sshd-session[51714]: Failed password for root from ::1 port 39470 ssh2
Feb 13 10:28:12 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:28:19 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:32:12 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:19 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:24 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
```

Check Successful Logins

`last`

```
(kali@kali)-[~]
└─$ last
kali      tty7          :0                Fri Feb 13 10:31 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:31 - 10:31 (00:00)
kali      pts/1         ::1               Fri Feb 13 10:28 - still logged in
kali      pts/1         ::1               Fri Feb 13 10:27 - 10:28 (00:00)
kali      tty7          :0                Fri Feb 13 10:27 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:26 - 10:27 (00:00)
kali      tty7          :0                Fri Feb 13 08:24 - 08:28 (00:04)
lightdm   tty7          :0                Fri Feb 13 08:24 - 08:24 (00:00)
kali      tty7          :0                Fri Feb 13 08:22 - still logged in
lightdm   tty7          :0                Fri Feb 13 08:22 - 08:22 (00:00)
kali      tty7          :0                Fri Feb 13 07:31 - still logged in
```

Correlate Events (Attack Chain)

Check timeline:

`sudo journalctl --since "10 minutes ago"`

```
(kali@kali)-[~]
$ sudo journalctl --since "10 minutes ago"
Feb 13 10:34:20 kali kernel: 15:33:08.468686 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Feb 13 10:34:21 kali kernel: 15:34:20.989041 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Feb 13 10:34:21 kali kernel: 15:34:20.998024 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Feb 13 10:34:22 kali kernel: 15:34:21.006539 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Feb 13 10:34:22 kali kernel: 15:34:22.102735 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Feb 13 10:34:22 kali kernel: 15:34:22.143849 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Feb 13 10:34:46 kali sudo[3255]: kali : TTYpts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl _COMM=sshd
Feb 13 10:34:46 kali sudo[3255]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Feb 13 10:34:47 kali sudo[3255]: pam_unix(sudo:session): session closed for user root
Feb 13 10:34:56 kali kernel: 15:34:22.146065 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
```

5. Contain the incident by isolating affected systems or accounts

The first step is to identify the affected systems or user accounts.

Compromised accounts should be immediately disabled or locked.

Passwords must be reset to prevent further unauthorized access.

If a system is infected, it should be disconnected from the network.

Network isolation prevents the attacker from moving laterally.

Access to shared drives and sensitive resources should be restricted.

Suspicious IP addresses can be blocked through firewall rules.

Active malicious sessions should be terminated immediately.

Backup systems should be checked to ensure they are not affected.

Monitoring should continue to confirm that the threat is contained.

Proper documentation of containment actions must be maintained.

Containment helps minimize damage and protect critical assets.

It is a crucial step before removing the threat completely.

Effective containment reduces business disruption and data loss.

Identify the Attacked User or IP (from Logs)

`sudo journalctl | grep "Failed password"`

```
(kali@kali)-[~]
$ sudo journalctl | grep "Failed password"
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 10 09:18:42 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:49 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:55 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:41:09 kali sshd-session[51714]: Failed password for root from ::1 port 39470 ssh2
Feb 13 10:28:12 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:28:19 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:32:12 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:19 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:24 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
```

Lock the Affected User Account (Immediate Containment)

Replace username with the real user:


```
sudo passwd -l testuser
```

```
(kali㉿kali)-[~]  
$ sudo passwd -l testuser  
passwd: password changed.
```

Step 3: Block the Attacker IP Address

If you saw an IP (example: 192.168.1.100):

Using UFW (if enabled):

```
sudo ufw deny from 192.168.1.100
```

```
(kali㉿kali)-[~]  
$ sudo ufw deny from 192.168.1.100  
Skipping adding existing rule
```

Reload firewall:

```
sudo ufw reload
```

```
(kali㉿kali)-[~]  
$ sudo ufw reload  
Firewall reloaded
```

Step 4 (Optional): Stop SSH Temporarily

If attack is ongoing:

```
sudo systemctl stop ssh
```

```
(kali㉿kali)-[~]  
$ sudo systemctl stop ssh
```

6. Remove the threat and fix the root cause

The first step is to identify the exact source of the compromise.

Malware or unauthorized files should be deleted using security tools.

Compromised accounts must have passwords reset and privileges reviewed.

Security patches and updates should be installed to fix vulnerabilities.

Misconfigured settings that allowed the attack must be corrected.

If necessary, reinstall or restore the system from a clean backup.

Remove any unauthorized user accounts created by the attacker.

Scan the entire system to ensure no hidden backdoors remain.

Check logs to confirm that malicious activity has stopped.

Root cause analysis should be performed to understand how the attack occurred.

Weak passwords, outdated software, or poor access control are common causes.

Implement stronger security controls to prevent recurrence.

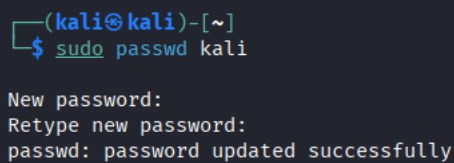
Document all removal and remediation actions taken.

Fixing the root cause ensures long-term protection and system stability.

Step 1: Reset Password of Affected Account

Replace kali with your real username if different:

sudo passwd kali

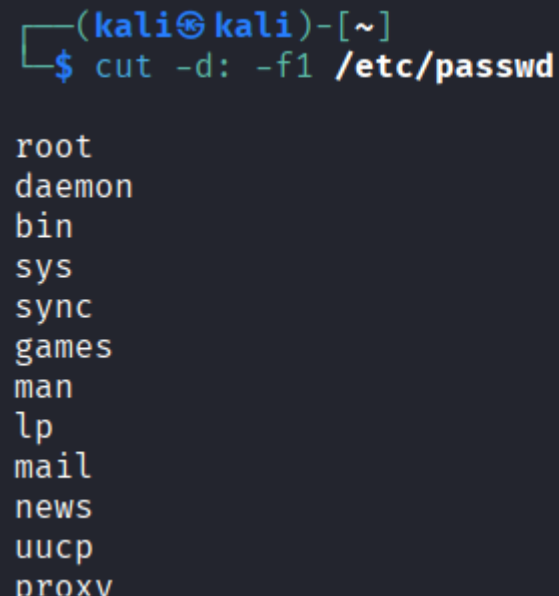


```
(kali@kali)-[~]  
$ sudo passwd kali  
  
New password:  
Retype new password:  
passwd: password updated successfully
```

Step 2: Check for Unknown Users (Remove If Found)

List users:

cut -d: -f1 /etc/passwd



```
(kali@kali)-[~]  
$ cut -d: -f1 /etc/passwd  
  
root  
daemon  
bin  
sys  
sync  
games  
man  
lp  
mail  
news  
uucp  
proxy
```

Step 3: Remove / Block Attacker IP (If Identified)

If logs showed an IP (example 192.168.1.100):

Using UFW:

sudo ufw deny from 192.168.1.100

sudo ufw reload

```
(kali㉿kali)-[~]  
$ sudo ufw deny from 192.168.1.100  
sudo ufw reload  
  
Skipping adding existing rule  
Firewall reloaded
```

Step 4: Update System (Fix Vulnerabilities)

sudo apt update

sudo apt upgrade -y

```
(kali㉿kali)-[~]  
$ sudo apt update  
sudo apt upgrade -y  
  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.7 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]  
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [117 kB]  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [271 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]  
Fetched 74.3 MB in 18s (4,213 kB/s)  
62 packages can be upgraded. Run 'apt list --upgradable' to see them.  
The following packages were automatically installed and are no longer required:
```

Step 5: Harden SSH (Root Cause Fix)

Edit SSH config:

sudo nano /etc/ssh/sshd_config

```
#AuthorizedKeysCommand none  
#AuthorizedKeysCommandUser nobody  
  
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts  
#HostbasedAuthentication no  
# Change to yes if you don't trust ~/.ssh/known_hosts for  
# HostbasedAuthentication  
#IgnoreUserKnownHosts no  
# Don't read the user's ~/.rhosts and ~/.shosts files  
#IgnoreRhosts yes  
  
# To disable tunneled clear text passwords, change to "no" here!  
#PasswordAuthentication yes  
#PermitEmptyPasswords no  
  
# Change to "yes" to enable keyboard-interactive authentication. Depending on  
# the system's configuration, this may involve passwords, challenge-response,  
# one-time passwords or some combination of these and other methods.  
# Beware issues with some PAM modules and threads.  
KbdInteractiveAuthentication no  
  
# Kerberos options  
#KerberosAuthentication no  
#KerberosOrLocalPasswd yes  
#KerberosTicketCleanup yes
```

Restart SSH:

sudo systemctl restart ssh

```
(kali㉿kali)-[~]  
$ sudo systemctl restart ssh
```

Step 6 (Recommended): Install Fail2Ban (Prevent Future Brute Force)

sudo apt install fail2ban -y

sudo systemctl enable fail2ban

sudo systemctl start fail2ban

```
(kali㉿kali)-[~]  
$ sudo apt install fail2ban -y  
sudo systemctl enable fail2ban  
sudo systemctl start fail2ban  
  
The following packages were automatically installed and are no longer required:  
  librubberband2  libsnmp40t64  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  fail2ban  
  
Installing dependencies:  
  python3-systemd  
  
Suggested packages:  
  mailx  system-log-daemon  monit
```

Check status:

sudo systemctl status fail2ban

```
(kali㉿kali)-[~]  
$ sudo systemctl status fail2ban  
  
● fail2ban.service - Fail2Ban Service  
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)  
   Active: active (running) since Sat 2026-02-14 08:00:46 EST; 57s ago  
  Invocation: 7d88a3a4834746b3815fbd558dee7200  
     Docs: man:fail2ban(1)  
  Main PID: 25250 (fail2ban-server)  
    Tasks: 5 (limit: 2118)  
   Memory: 14.2M (peak: 14.7M)  
      CPU: 326ms  
   CGroup: /system.slice/fail2ban.service  
           └─25250 /usr/bin/python3 /usr/bin/fail2ban-server -xf start  
  
Feb 14 08:00:46 kali systemd[1]: Started fail2ban.service - Fail2Ban Service.  
Feb 14 08:00:46 kali fail2ban-server[25250]: Server ready
```

7. Restore systems to a secure state

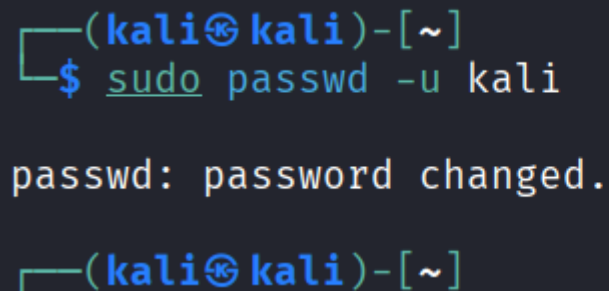
8. The first step is to verify that the threat has been completely removed. Systems may be restored from clean and verified backups if necessary. All security patches and updates should be applied before reconnecting to the network.

User passwords should be reset to ensure account security.
Access permissions must be reviewed and corrected based on least privilege.
Security configurations such as firewalls and antivirus should be re-enabled.
System integrity should be checked to confirm no files were altered.
Test the system to ensure all services are functioning properly.
Monitor logs closely for any signs of recurring suspicious activity.
Inform users when systems are safely restored.
Ensure backup systems are updated and secure.
Document the restoration process for future reference.
Conduct a final security review before full production use.
Proper restoration ensures business continuity and prevents further compromise.

Step 1: Unlock Legitimate User Account (If Previously Locked)

If you locked your account earlier (example: kali):

`sudo passwd -u kali`



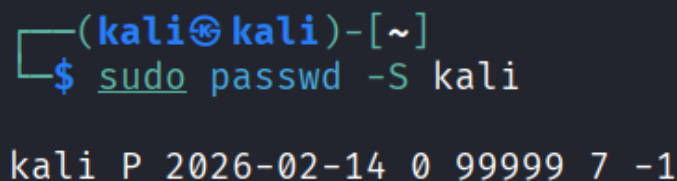
```
(kali㉿kali)-[~]  
$ sudo passwd -u kali  
  
passwd: password changed.  
  
(kali㉿kali)-[~]
```

Verify:

`sudo passwd -S kali`

You should see:

kali P ...



```
(kali㉿kali)-[~]  
$ sudo passwd -S kali  
  
kali P 2026-02-14 0 99999 7 -1
```

Step 2: Restart Essential Services (SSH)

If SSH was stopped during containment:

`sudo systemctl start ssh`

```
kali@kali:~$ sudo systemctl start ssh
```

Check status:

sudo systemctl status ssh

```
(kali@kali)-[~]
$ sudo systemctl status ssh

● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-02-14 07:59:47 EST; 5min ago
     Invocation: 01aa69d841ff41c8b6839c3e1019b567
       Docs: man:sshd(8)
             man:sshd_config(5)
    Main PID: 24090 (sshd)
      Tasks: 1 (limit: 2118)
     Memory: 2.8M (peak: 3.5M)
        CPU: 40ms
    CGroup: /system.slice/ssh.service
            └─24090 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Step 3: Ensure Firewall Is Enabled

sudo ufw enable

sudo ufw status

```
(kali@kali)-[~]
$ sudo ufw enable
sudo ufw status
```

```
Firewall is active and enabled on system startup
Status: active
```

To	Action	From
--	---	---
22	ALLOW	Anywhere
80	ALLOW	Anywhere
443	ALLOW	Anywhere
Anywhere	DENY	192.168.10.3
22	ALLOW	192.168.1.10
80/tcp	ALLOW	Anywhere
23	DENY	Anywhere
21/tcp	DENY	Anywhere
Anywhere	DENY	192.168.1.50
Anywhere	DENY	192.168.1.100
22/tcp	ALLOW	Anywhere
22 (v6)	ALLOW	Anywhere (v6)
80 (v6)	ALLOW	Anywhere (v6)
443 (v6)	ALLOW	Anywhere (v6)
80/tcp (v6)	ALLOW	Anywhere (v6)
23 (v6)	DENY	Anywhere (v6)
21/tcp (v6)	DENY	Anywhere (v6)
22/tcp (v6)	ALLOW	Anywhere (v6)

Step 4: Verify System Is Fully Updated

```
sudo apt update
```

```
sudo apt upgrade -y
```

```
(kali@kali)-[~]
$ sudo apt update
sudo apt upgrade -y
```

```
Hit:1 http://http.kali.org/kali kali-rolling InRelease
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  librubberband2 libsnmp40t64
Use 'sudo apt autoremove' to remove them.
```

```
Not upgrading:
  colord libcolorhug2 libgbm1 libglx-mesa0 libgtk-4-bin mesa-libgallium python-tables-data
  graphviz libegl-mesa0 libgl1-mesa-dri libgtk-4-1 libgtkmm-4.0-0 nodejs python3-pygraphviz
```

```
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
```

```
(kali@kali)-[~]
```

Step 5: Confirm Fail2Ban Is Running (If Installed)

sudo systemctl status fail2ban

```
(kali㉿kali)-[~]
$ sudo systemctl status fail2ban

● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-02-14 08:00:46 EST; 6min ago
  Invocation: 7d88a3a4834746b3815fbd558dee7200
     Docs: man:fail2ban(1)
    Main PID: 25250 (fail2ban-server)
       Tasks: 5 (limit: 2118)
      Memory: 14.4M (peak: 16M)
         CPU: 1.266s
      CGroup: /system.slice/fail2ban.service
             └─25250 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Feb 14 08:00:46 kali systemd[1]: Started fail2ban.service - Fail2Ban Service.
Feb 14 08:00:46 kali fail2ban-server[25250]: Server ready
```

Step 6: Monitor Logs for Any New Suspicious Activity

sudo journalctl | grep "Failed"

```
(kali㉿kali)-[~]
$ sudo journalctl | grep "Failed"

Nov 26 23:23:41 kali wireplumber[964]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NoReply: The remote
Nov 26 23:23:45 kali lightdm[930]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:23:46 kali lightdm[1034]: Failed to write utmpx: No such file or directory
Nov 26 23:23:46 kali wireplumber[1074]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner: No such name
Nov 26 23:25:23 kali wireplumber[743]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner: No such name
Nov 26 23:25:28 kali lightdm[707]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:25:28 kali lightdm[810]: Failed to write utmpx: No such file or directory
Nov 26 23:25:28 kali wireplumber[849]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner: No such name
Nov 27 08:38:12 kali wireplumber[839]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner: No such name
Nov 27 08:38:16 kali lightdm[804]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 27 08:38:17 kali lightdm[908]: Failed to write utmpx: No such file or directory
Nov 27 08:38:17 kali wireplumber[947]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner: No such name
```

and:

last

```
(kali㉿kali)-[~]
$ last

kali      tty7          :0                Sat Feb 14 07:46 - still logged in
lightdm   tty7          :0                Sat Feb 14 07:46 - 07:46 (00:00)
kali      tty7          :0                Sat Feb 14 07:26 - 07:33 (00:06)
lightdm   tty7          :0                Sat Feb 14 07:26 - 07:26 (00:00)
kali      tty7          :0                Fri Feb 13 10:31 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:31 - 10:31 (00:00)
kali      pts/1         ::1               Fri Feb 13 10:28 - still logged in
kali      pts/1         ::1               Fri Feb 13 10:27 - 10:28 (00:00)
kali      tty7          :0                Fri Feb 13 10:27 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:26 - 10:27 (00:00)
kali      tty7          :0                Fri Feb 13 08:24 - 08:28 (00:04)
```

9. Document the incident timeline and actions taken.

Documenting the incident timeline means recording all events related to the security incident in chronological order.

The timeline should begin with the date and time the incident was first detected.

Include details about how the incident was identified, such as log alerts or user reports.

Record all suspicious activities observed during investigation.

Note the systems, accounts, and data affected by the incident.

Document the classification of the incident and its severity level.

List all containment actions taken, such as isolating systems or disabling accounts.

Include details of threat removal and remediation steps performed.

Record when systems were restored to normal operation.

Mention any communication made to management or stakeholders.

Attach relevant log files, screenshots, or evidence for reference.

Ensure all actions are clearly explained and time-stamped.

Documentation should be accurate, clear, and professional.

This record helps in legal, audit, and compliance requirements.

Proper documentation also improves future incident response planning.

Step 1: Extract Failed Login Events with Timestamps

`sudo journalctl | grep "Failed password"`

```
(kali@kali)-[~]
└─$ sudo journalctl | grep "Failed password"

[sudo] password for kali:
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 10 09:18:42 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:49 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:18:55 kali sshd-session[27579]: Failed password for root from ::1 port 39056 ssh2
Feb 10 09:41:09 kali sshd-session[51714]: Failed password for root from ::1 port 39470 ssh2
Feb 13 10:28:12 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:28:19 kali sshd-session[2449]: Failed password for kali from ::1 port 57760 ssh2
Feb 13 10:32:12 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:19 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
Feb 13 10:32:24 kali sshd-session[2003]: Failed password for kali from ::1 port 34540 ssh2
```

Step 2: Check Successful Logins (If Any)

`last`

```
(kali@kali)-[~]
└─$ last
kali      tty7          :0                Sun Feb 15 06:10 - still logged in
lightdm   tty7          :0                Sun Feb 15 06:07 - 06:10 (00:03)
kali      tty7          :0                Sat Feb 14 07:46 - still logged in
lightdm   tty7          :0                Sat Feb 14 07:46 - 07:46 (00:00)
kali      tty7          :0                Sat Feb 14 07:26 - 07:33 (00:06)
lightdm   tty7          :0                Sat Feb 14 07:26 - 07:26 (00:00)
kali      tty7          :0                Fri Feb 13 10:31 - still logged in
lightdm   tty7          :0                Fri Feb 13 10:31 - 10:31 (00:00)
kali      pts/1          ::1               Fri Feb 13 10:28 - still logged in
kali      pts/1          ::1               Fri Feb 13 10:27 - 10:28 (00:00)
```

Step 3: Review Containment Actions (Service Stops / Account Locks)

Check SSH service history:

`sudo journalctl -u ssh`


```
(kali@kali)-[~]
$ sudo journalctl -u ssh
Feb 03 10:45:42 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 03 10:45:42 kali sshd[3506]: Server listening on 0.0.0.0 port 22.
Feb 03 10:45:42 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 03 10:45:42 kali sshd[3506]: Server listening on :: port 22.
Feb 03 10:46:46 kali sshd-session[3835]: Connection closed by 127.0.0.1 port 49538 [preauth]
Feb 03 10:47:43 kali sshd-session[4072]: Connection closed by 127.0.0.1 port 33820 [preauth]
Feb 03 10:49:11 kali unix_chkpwd[5166]: password check failed for user (kali)
Feb 03 10:49:11 kali sshd-session[5036]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=kali
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): request_smbloginuser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9):
Feb 03 10:49:11 kali sshd-session[5036]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'kali')
Feb 03 10:49:12 kali sshd-session[5036]: Failed password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: Accepted password for kali from 127.0.0.1 port 59610 ssh2
Feb 03 10:49:59 kali sshd-session[5036]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
```

Step 4: Verify Firewall or Fail2Ban Actions (If Used)

For UFW:

sudo ufw status numbered

```
(kali@kali)-[~]
$ sudo ufw status numbered

Status: active

    To Action From
    --
[ 1] 22 ALLOW IN Anywhere
[ 2] 80 ALLOW IN Anywhere
[ 3] 443 ALLOW IN Anywhere
[ 4] Anywhere DENY IN 192.168.10.3
[ 5] 22 ALLOW IN 192.168.1.10
[ 6] 80/tcp ALLOW IN Anywhere
[ 7] 23 DENY IN Anywhere
[ 8] 21/tcp DENY IN Anywhere
[ 9] Anywhere DENY IN 192.168.1.50
[10] Anywhere DENY IN 192.168.1.100
[11] 22/tcp ALLOW IN Anywhere
[12] 22 (v6) ALLOW IN Anywhere (v6)
[13] 80 (v6) ALLOW IN Anywhere (v6)
[14] 443 (v6) ALLOW IN Anywhere (v6)
[15] 80/tcp (v6) ALLOW IN Anywhere (v6)
[16] 23 (v6) DENY IN Anywhere (v6)
[17] 21/tcp (v6) DENY IN Anywhere (v6)
[18] 22/tcp (v6) ALLOW IN Anywhere (v6)
```

For Fail2Ban:

sudo fail2ban-client status

```
(kali@kali)-[~]
$ sudo fail2ban-client status

Status
├─ Number of jail:      1
└─ Jail list:  sshd
```

Step 5: Save Evidence (Optional but Recommended)

Export logs:

sudo journalctl > incident_logs.txt

```
(kali@kali)-[~]
$ sudo journalctl > incident_logs.txt
```

10 Recommend preventive security improvements

Recommending preventive security improvements helps reduce the risk of future incidents.

Strong password policies should be implemented to prevent brute-force attacks.

Multi-factor authentication (MFA) should be enabled for all critical accounts.

Regular system updates and security patching must be performed.

Firewalls and intrusion detection systems should be properly configured.

User access should follow the principle of least privilege.

Account lockout policies can prevent repeated failed login attempts.

Regular log monitoring and security audits should be conducted.

Employees should receive cybersecurity awareness training.

Sensitive data should be encrypted to prevent unauthorized access.

Backup systems must be maintained and tested regularly.

Network segmentation can limit the spread of attacks.

Security policies and procedures should be reviewed periodically.

Vulnerability assessments and penetration testing should be performed.

Implementing these measures strengthens overall organizational security posture.

1. Keep System Updated (Patch Vulnerabilities)

`sudo apt update`

`sudo apt upgrade -y`

```
(kali@kali)~$ sudo apt update
sudo apt upgrade -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  librubberband2 libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Not upgrading:
colora libcolorhug2 libgbm1 libglx-mesa0 libgtk-4-bin mesa-libgallium python-tables-data python3-tables yelp
graphviz libegl-mesa0 libgl1-mesa-dri libgtk-4-1 libgtkmm-4.0-0 nodejs python3-pygraphviz python3-tables-lib

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
```

2. Enforce Strong Passwords

Edit PAM password policy:

`sudo nano /etc/security/pwquality.conf`

Set (or confirm):

`minlen = 10`

`ucredit = -1`

`lcredit = -1`

dcredit = -1

ocredit = -1

```
upgrading: 0; installing: 0; removing: 0; not upgrading: 0  
(kali㉿kali)-[~]  
$ sudo nano /etc/security/pwquality.conf
```

3. Disable Root Login via SSH

sudo nano /etc/ssh/sshd_config

```
(kali㉿kali)-[~]  
$ sudo nano /etc/ssh/sshd_config
```

Restart SSH:

sudo systemctl restart ssh

```
(kali㉿kali)-[~]  
$ sudo systemctl restart ssh
```

4. Enable Firewall (UFW)

sudo ufw enable

sudo ufw allow ssh

sudo ufw status

```
(kali㉿kali)-[~]
└─$ sudo ufw enable
sudo ufw allow ssh
sudo ufw status

Firewall is active and enabled on system startup
Skipping adding existing rule
Skipping adding existing rule (v6)
Status: active
```

To	Action	From
22	ALLOW	Anywhere
80	ALLOW	Anywhere
443	ALLOW	Anywhere
Anywhere	DENY	192.168.10.3
22	ALLOW	192.168.1.10
80/tcp	ALLOW	Anywhere
23	DENY	Anywhere
21/tcp	DENY	Anywhere
Anywhere	DENY	192.168.1.50
Anywhere	DENY	192.168.1.100
22/tcp	ALLOW	Anywhere
22 (v6)	ALLOW	Anywhere (v6)
80 (v6)	ALLOW	Anywhere (v6)
443 (v6)	ALLOW	Anywhere (v6)
80/tcp (v6)	ALLOW	Anywhere (v6)
23 (v6)	DENY	Anywhere (v6)
21/tcp (v6)	DENY	Anywhere (v6)
22/tcp (v6)	ALLOW	Anywhere (v6)

5. Install Fail2Ban (Auto-block Brute Force)

`sudo apt install fail2ban -y`

`sudo systemctl enable fail2ban`

`sudo systemctl start fail2ban`

```
(kali㉿kali)-[~]
└─$ sudo apt install fail2ban -y
sudo systemctl enable fail2ban
sudo systemctl start fail2ban

fail2ban is already the newest version (1.1.0-9).
The following packages were automatically installed and are no longer required:
  librubberband2 libsnmp40t64
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
  Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
  Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
```

Check:

`sudo systemctl status fail2ban`

```
(kali㉿kali)-[~]
└─$ sudo systemctl status fail2ban

● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-02-15 06:07:24 EST; 13min ago
 Invocation: 07c9ff0ee0a84d3ead0c1c640476d9e5
    Docs: man:fail2ban(1)
   Main PID: 870 (fail2ban-server)
     Tasks: 5 (limit: 2118)
  Memory: 17.5M (peak: 31.7M, swap: 496K, swap peak: 496K)
     CPU: 3.665s
    CGroup: /system.slice/fail2ban.service
            └─870 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Feb 15 06:07:24 kali systemd[1]: Started fail2ban.service - Fail2Ban Service.
Feb 15 06:07:29 kali fail2ban-server[870]: Server ready
```

6. Remove Unnecessary Users

cut -d: -f1 /etc/passwd

```
(kali@kali)-[~]
$ cut -d: -f1 /etc/passwd

root
daemon
bin
sys
sync
```

7. Regular Log Monitoring

sudo journalctl | grep "Failed"

```
(kali@kali)-[~]
$ sudo journalctl | grep "Failed"

Nov 26 23:23:41 kali wireplumber[964]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:23:45 kali lightdm[930]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:23:46 kali lightdm[1034]: Failed to write utmpx: No such file or directory
Nov 26 23:23:46 kali wireplumber[1074]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:23 kali wireplumber[743]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 26 23:25:28 kali lightdm[707]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 26 23:25:28 kali lightdm[810]: Failed to write utmpx: No such file or directory
Nov 26 23:25:28 kali wireplumber[849]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 27 08:38:12 kali wireplumber[839]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 27 08:38:16 kali lightdm[804]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
Nov 27 08:38:17 kali lightdm[908]: Failed to write utmpx: No such file or directory
Nov 27 08:38:17 kali wireplumber[947]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Nov 29 09:04:45 kali wireplumber[824]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
```

Conclusion :-

The incident response process begins with simulating a basic security incident.

This helps in understanding how attacks like failed logins or unauthorized access occur.

System and authentication logs are then analyzed to detect suspicious activity.

Unusual login attempts or privilege misuse indicate possible threats.

The incident is classified based on attack type and severity level.

This helps determine the impact and response priority.

Next, the incident is contained by isolating affected systems or accounts.

Compromised accounts are disabled and infected systems are disconnected.

After containment, the threat is removed completely.

The root cause of the incident is identified and fixed.

Systems are restored using clean backups and security patches.

Passwords and access controls are strengthened.

All actions and timelines are properly documented.
Preventive security measures are recommended and implemented.
This structured approach ensures better security and future protection.